Comparing regression models for the relationship between blood glucose concentration and insulin concentration between diabetic and non-diabetic cohorts

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Introduction

Diabetes mellitus is a rapidly growing global health concern characterised by chronic dysregulation of blood glucose concentration as a result of impaired insulin production, reduced insulin sensitivity, or a combination of both. The increasing prevalence of diabetes over recent decades, particularly in developing nations, can be attributed to rapid lifestyle changes and dietary transitions. According to World Health Organization estimates discussed by *Ramachandran and Snehalatha (2009)*, India has experienced one of the fastest increases in diabetes prevalence worldwide,highlighting the importance of the Indian population as a subject of metabolic research. This growing burden has motivated continued investigation into the physiological mechanisms governing glucose regulation and insulin response.

Insulin plays a central role in maintaining glucose homeostasis by facilitating cellular glucose uptake and regulating metabolic storage pathways. However, insulin resistance and impaired pancreatic β-cell function disrupt this regulatory system, contributing to abnormal glucose accumulation and progression toward Type 2 diabetes mellitus *(Kahn et al., 2006)*. As noted by *Shoelson et al. (2006)*, this dysregulation is often accompanied by broader inflammatory and metabolic disturbances, further complicating prediction of insulin response across diabetic populations.

Given the complexity of these physiological interactions, statistical modelling provides a useful foundation for inferring measurable relationships between medical variables. This investigation explores the relationship between blood glucose concentration and insulin concentration using regression modelling techniques applied to a diabetes dataset. Particular emphasis was placed on determining whether subdivision of patients by diabetic status improved predictive performance relative to increasing model complexity.

Dataset & Methods

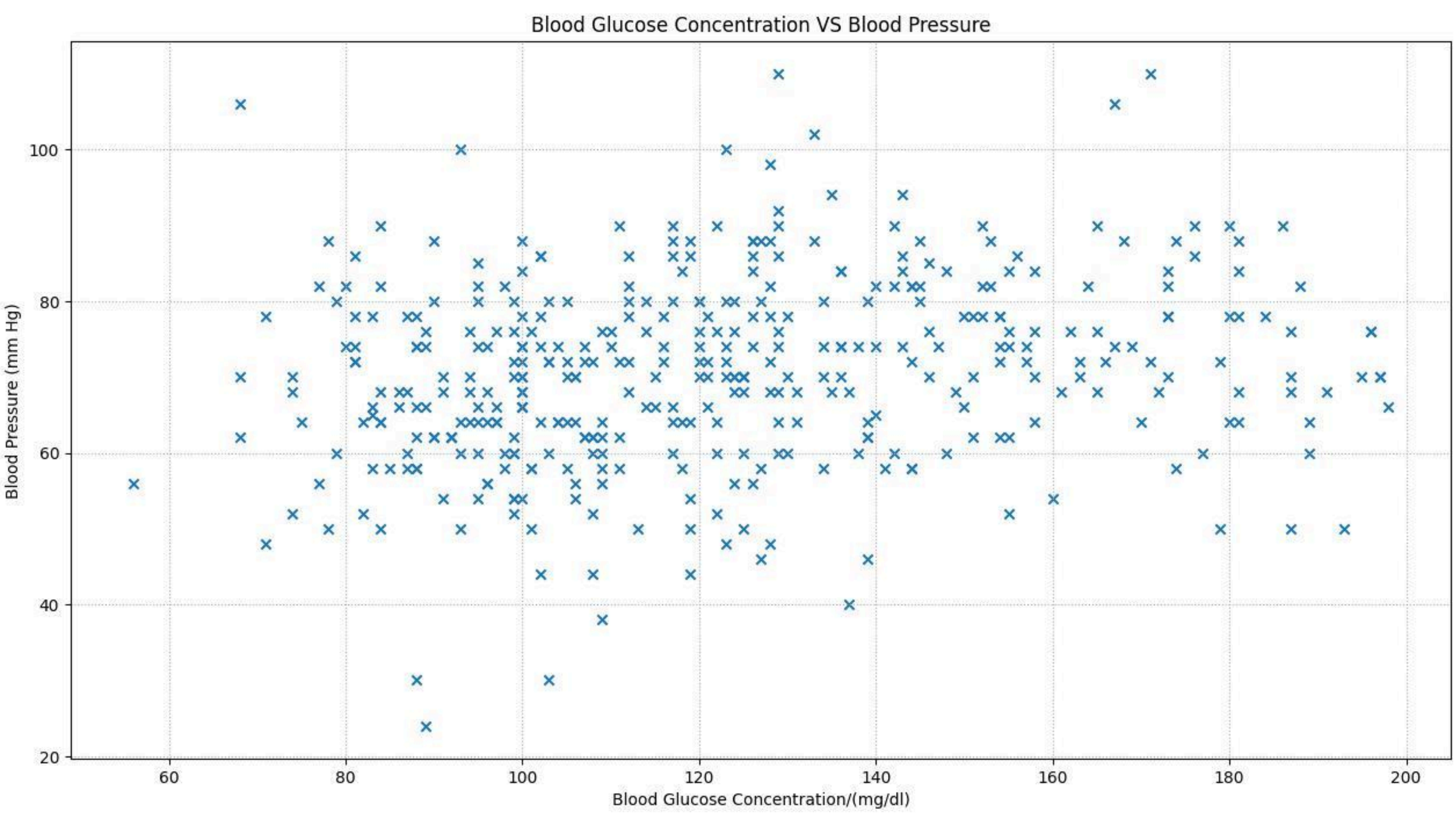
The dataset for this project was obtained from Kaggle (Khare, n.d.) and was originally sourced from the National Institute of Diabetes and Digestive and Kidney Diseases. The original dataset was a comprehensive collection of measurements to test correlations between diabetes prevalence and a variety of different physiological factors. The Kaggle database in particular focuses on female patients above the age of 21 years old and coming from Pima Indian heritage which creates a demographic consistency in the sample size being studied and helps limit the possible explanations for the observations made later in the paper.

This paper originally began as an exploratory study to determine which physiological factors correlated with blood glucose concentration most strongly. After initial plotting and modelling (discussed later) it became clear that the strongest predictive performance existed between blood glucose concentration and insulin concentration, particularly when the methodology of this investigation shifted from studying a global dataset to segregating the data points for diabetic and non-diabetic patients.

The dataset contained several measurements, particularly zero measurements for medical predictor variables such as blood glucose concentration and insulin concentration which hinted towards invalid or unavailable data. To decrease variability within the graphs and to conceptualise the relationships more clearly, these measurements were excluded from the analysis.

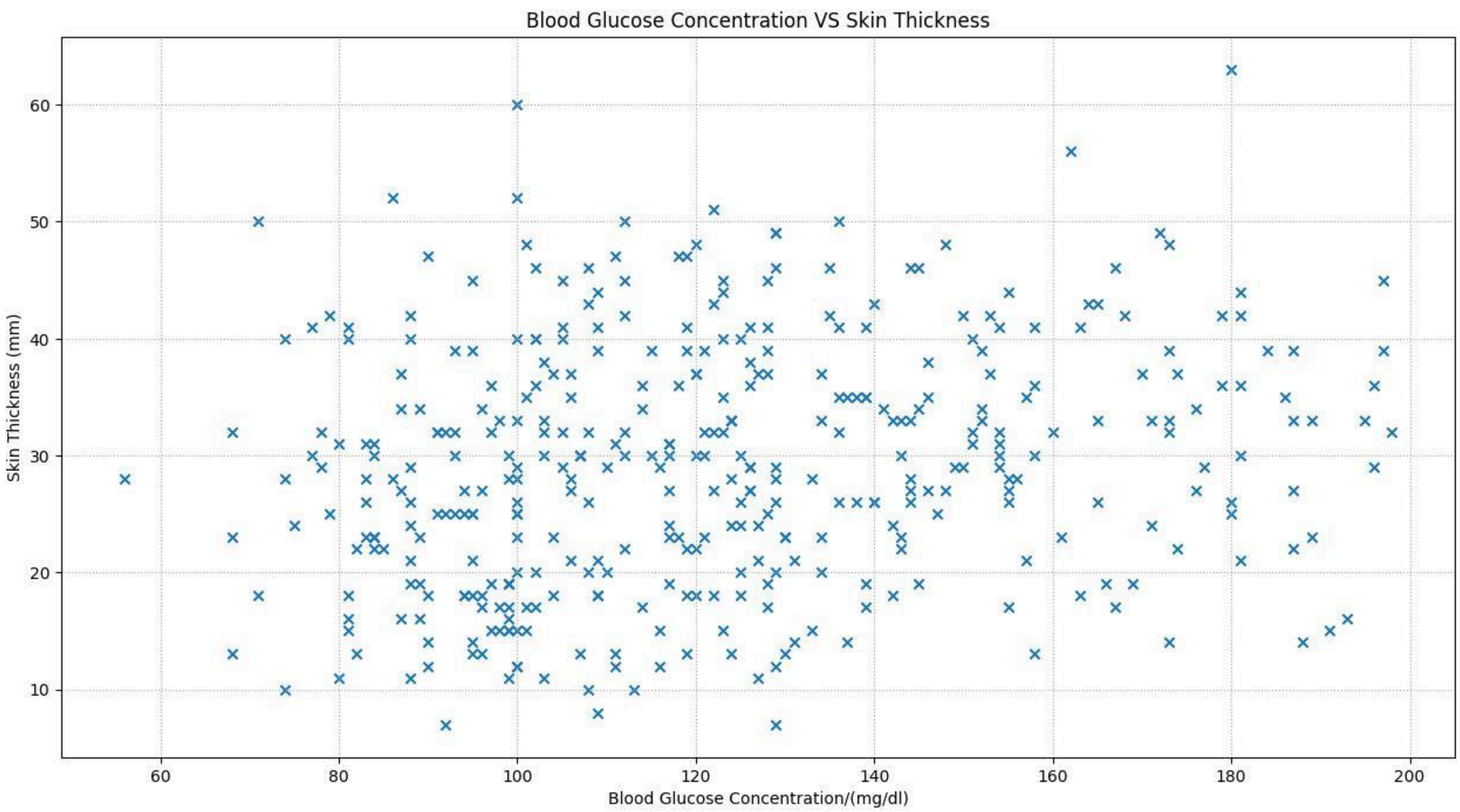
Regression models were utilised to determine a relationship between the two variables under investigation; models between global and sub-divided datasets were compared, and models utilising polynomial expressions of higher degree were also compared to determine which of these had a greater statistical significance.

Exploratory Analysis



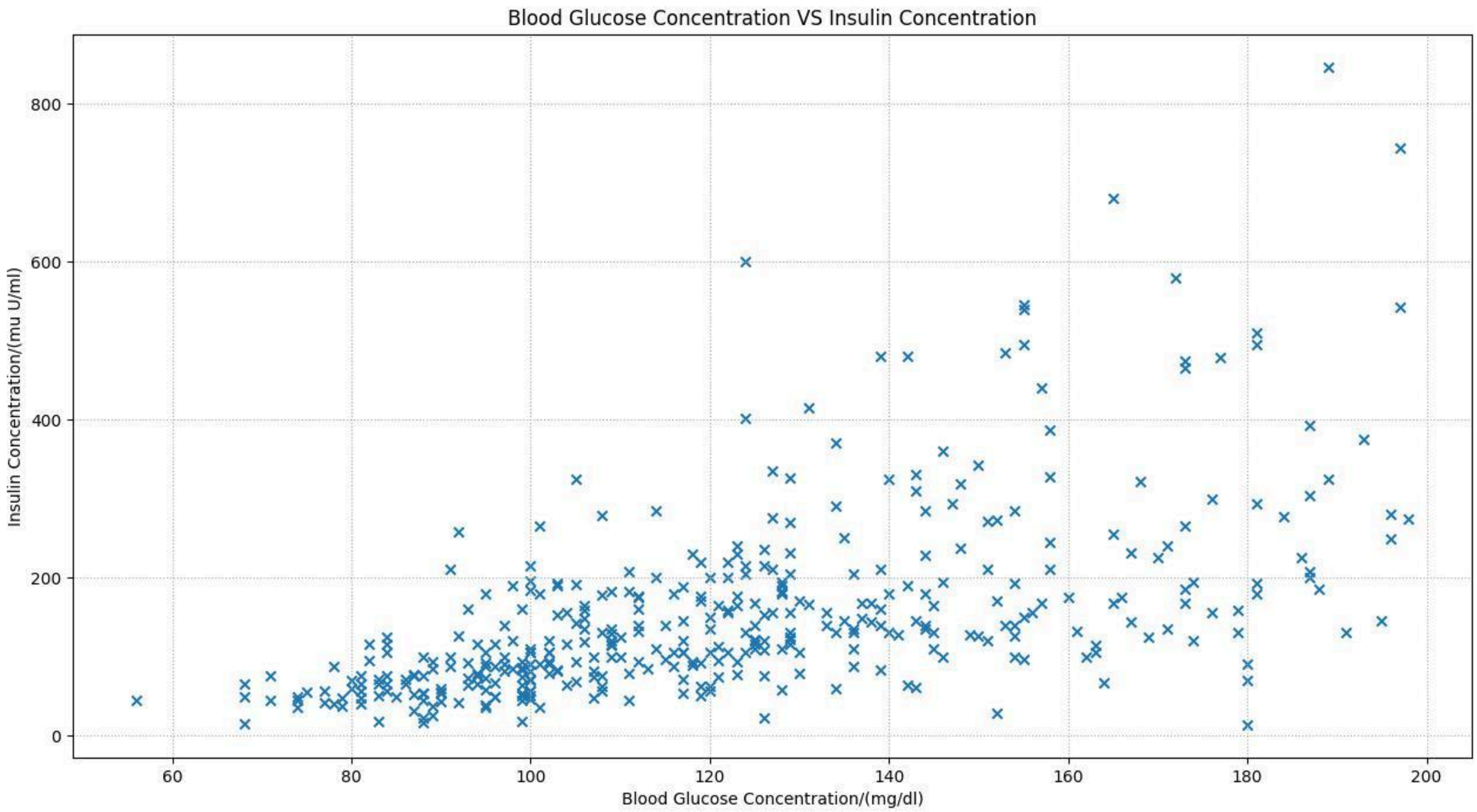
Graph 1. Correlation between Blood Glucose Concentration and Blood Pressure.

Visually the above scatter plot shows a weak correlation between the two variables. The data appears to follow no obvious trend, as the majority of patients were observed to have a Blood pressure between 60 and 80 mm Hg across the range of Blood Glucose concentrations.



Graph 2. Correlation between Blood Glucose Concentration and Skin Thickness.

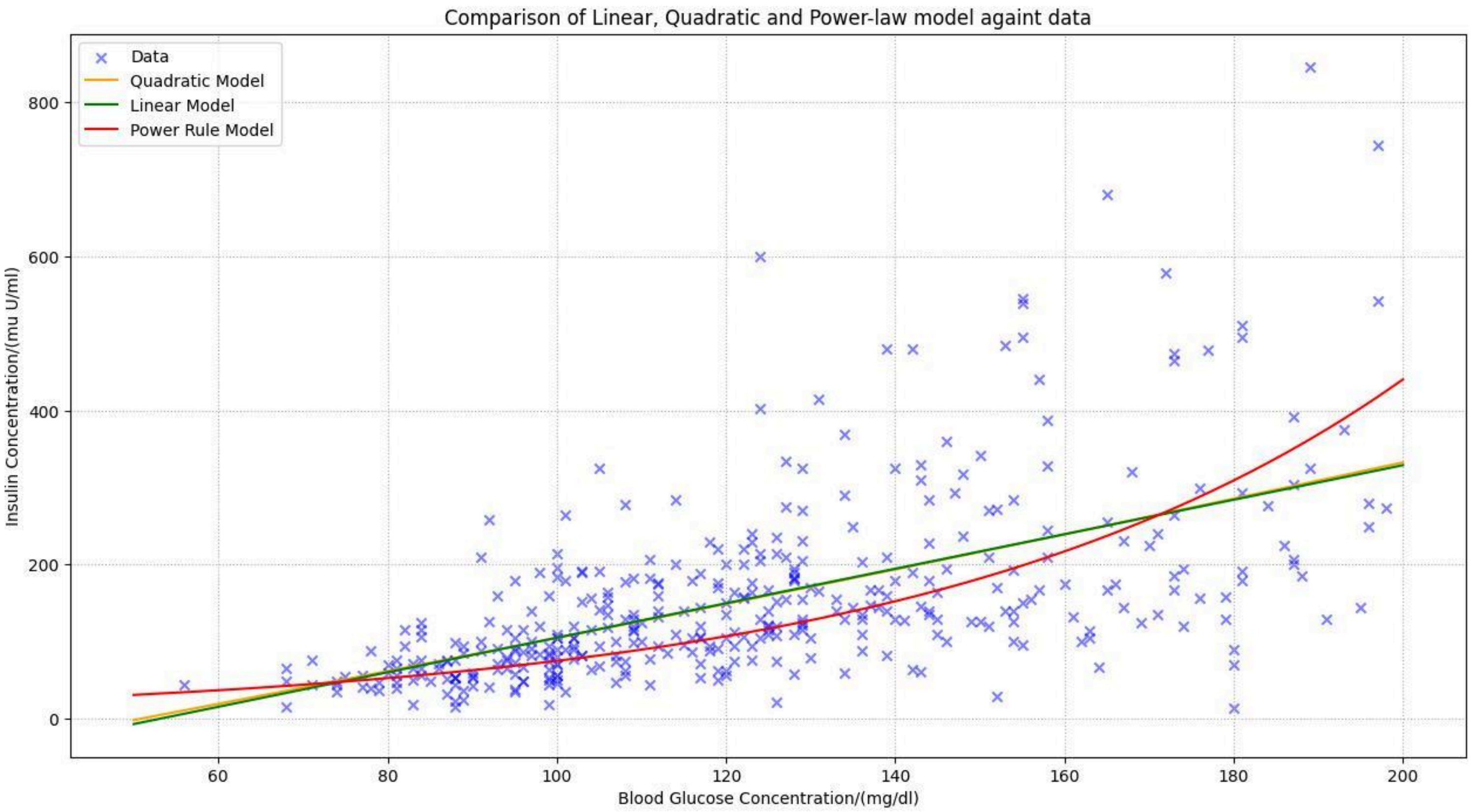
Visually the data appears to be randomly dispersed, with no functional relationship defining predictive performance



Graph 3. Correlation between Blood Glucose Concentration and Insulin Concentration.

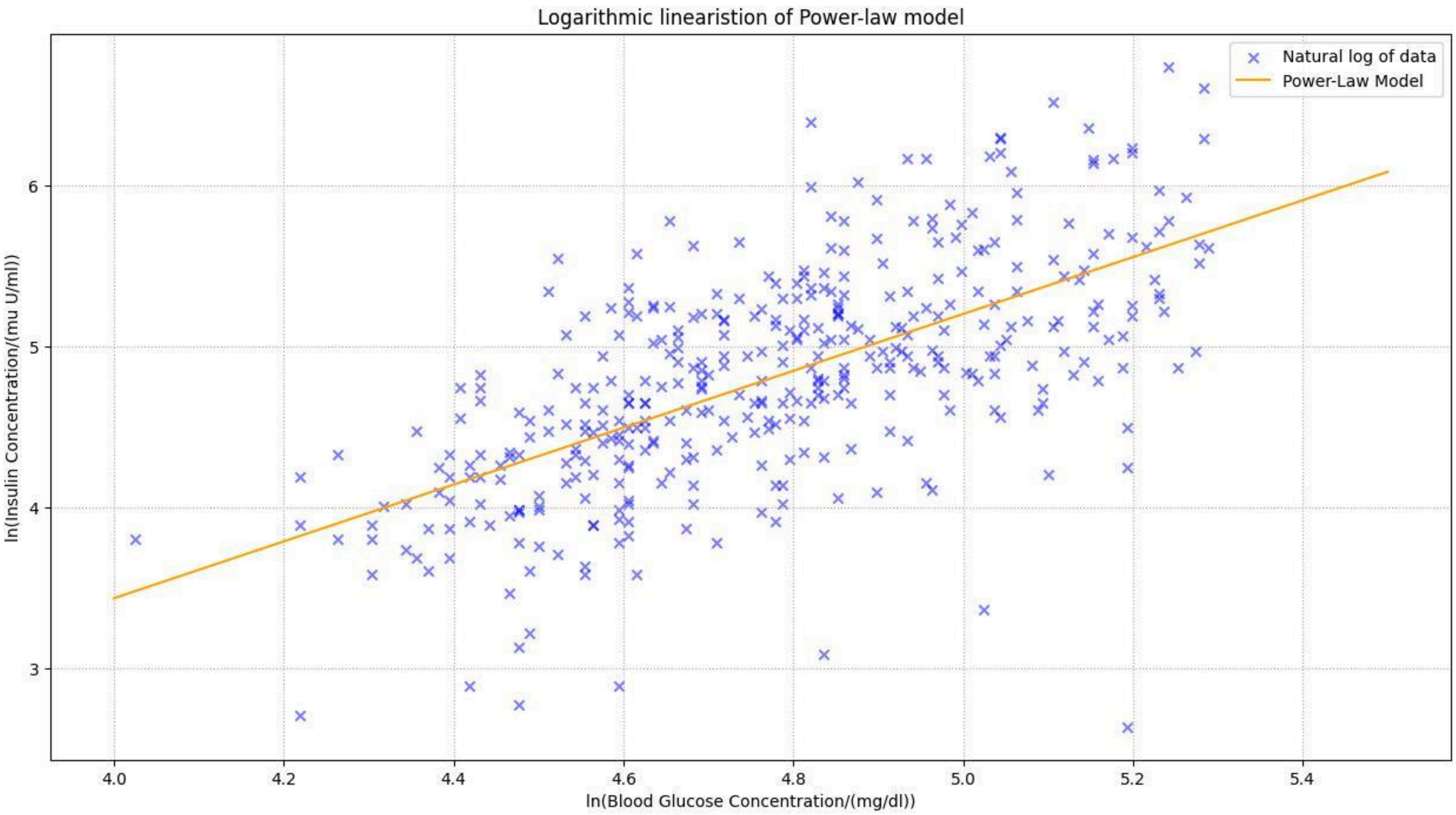
The above graph exhibited the strongest positive association among the physiological variables investigated. The relationship appeared approximately linear at lower glucose concentrations, although slight nonlinear curvature became visible at higher glucose levels. Additionally there appear to be several high Insulin concentration outliers.

Model Comparison



Graph 4. Comparing linear, quadratic and power-law regression models against the global dataset.

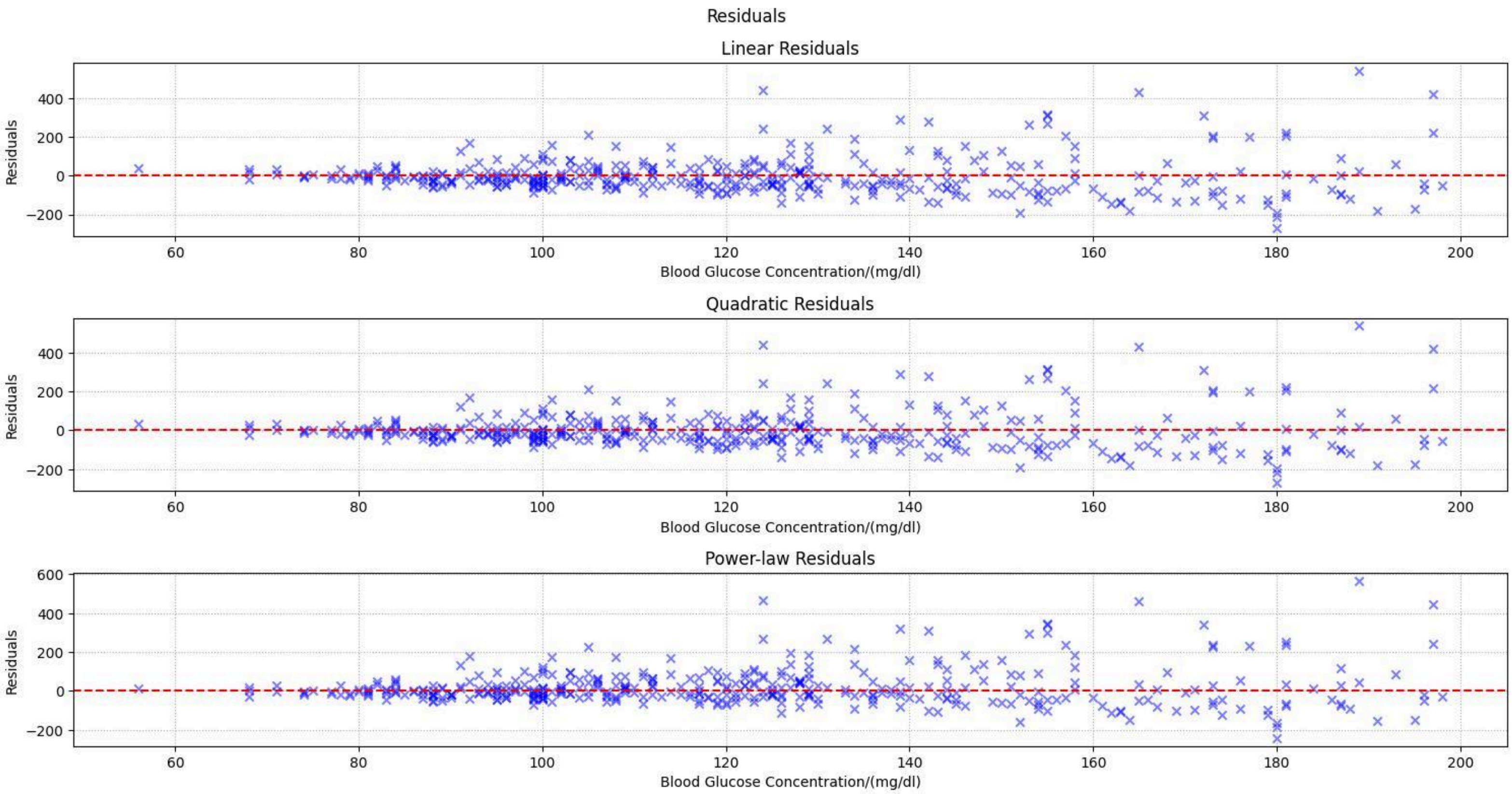
The linear model best fits the association between blood glucose concentration and insulin concentration. The quadratic model visually indicates that curvature has a negligible improvement to the relationship between the variables. The power-law model at first glance seems to best explain the relationship between the variables at high blood glucose concentrations; to be able to better visualise this, the model was isolated and logarithmically linearised as shown below.



Graph 5. Linearisation of Power-law model using logarithms.

A moderate positive relationship was observed between the investigated variables. The fit of the linearised model seems to support the non-linear relationship followed by the raw data points. This can be verified through statistical analysis and comparison of SSE and coefficient of determination (R^2) as done later in the paper.

Residual Analysis

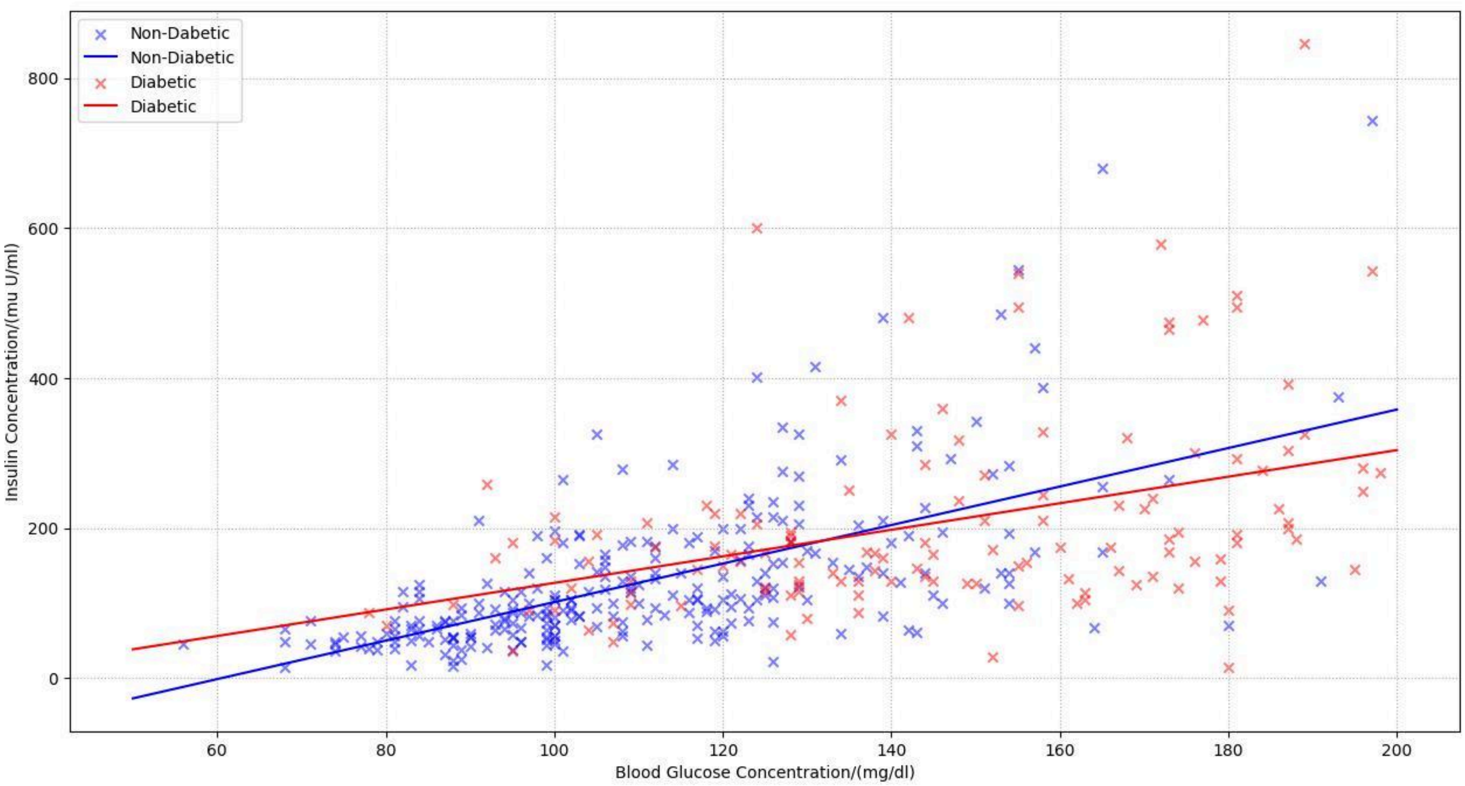


Graph 6. Plot of residuals against blood glucose concentration for the linear, quadratic and power-law models of the global dataset.

Increasing heteroscedasticity of the residuals was observed at higher blood glucose concentrations. This indicates the limitations of the predictive reliability of the models at higher blood glucose concentrations. This conclusion is further strengthened by the presence of outliers that deviate from the general trend with several insulin concentrations above 400 mu U/ml at high blood glucose concentrations as shown in *Graph 4.*

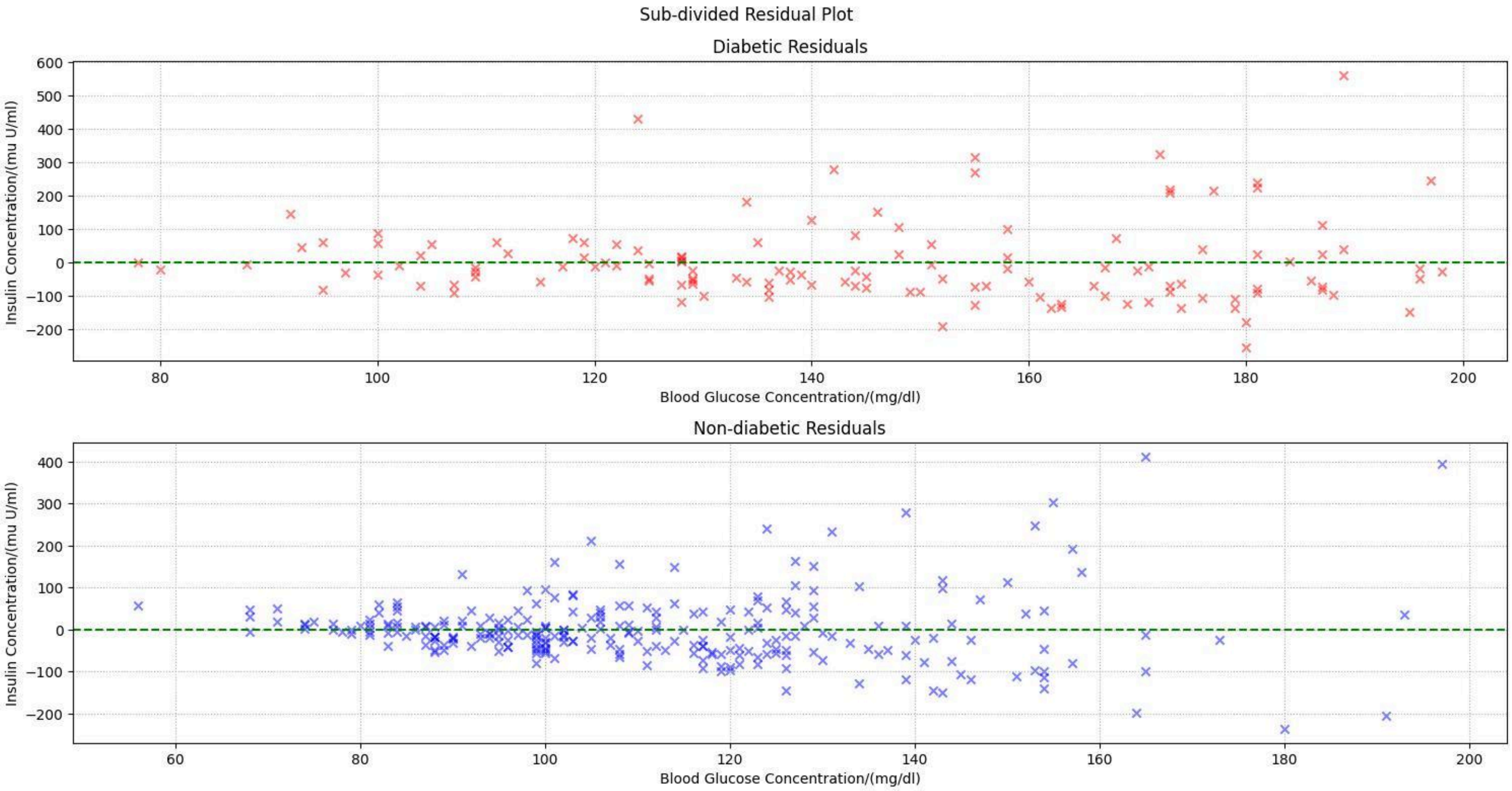
Subgroup Analysis

The decrease in model accuracy at high blood glucose concentrations indicated possible structural differences within the global dataset. This shifted the purpose of the paper towards identifying this discrepancy, thus the global dataset was sub-divided to segregate data points for diabetic and non-diabetic patients as shown below.



Graph 7. Separated scatter plots of measurements for diabetic and non-diabetic patients alongside accompanying linear regression models.

The general trend for both sub-groups indicates a moderate positive trend, however it was observed that the majority of outliers belonged to the diabetic group. To compare the variance of the points, residual plots for the individual sub-groups can be compared as done below.



Graph 8. Separated residual plots for linear regression models of both the diabetic and non-diabetic cohort.

Residual analysis of the subdivided datasets revealed substantial differences in model performance between diabetic and non-diabetic cohorts. Residuals within the non-diabetic subgroup were more tightly clustered around zero, indicating improved predictive consistency and reduced unexplained variance. In contrast, the diabetic subgroup exhibited greater residual dispersion and a broader spread of extreme deviations, suggesting increased physiological variability not fully captured by the regression model. These findings support the conclusion that subgroup separation improved model fit, while also highlighting the greater complexity of modelling insulin response within diabetic patients. While heteroscedastic behaviour persisted in both subgroups, it was less pronounced within the non-diabetic cohort.

Table of Results

Dataset	Model	SSE (4 s.f.)	R^2 (4 s.f.)
Global	Linear	3,660,000	0.3378
	Quadratic	3,660,000	0.3378
	Power-Law	3,864,000	0.3009
Diabetic	Linear	1,912,000	0.1584
	Quadratic	1,903,000	0.1622
Non-diabetic	Linear	1,706,000	0.3798
	Quadratic	1,687,000	0.3866

Table 1. Comparison of regression models between global and sub-divided datasets using SSE and coefficient of determination (R^2).

The results indicate that sub-group separation by diabetic status improved predictability between blood glucose concentration and insulin concentration over model complexity. While higher order models yielded negligible improvements within the global dataset, subgroup analysis produced noticeably improved R^2 values, particularly within the non-diabetic cohort.

Discussion and Conclusion

The results of this investigation suggest that physiological subgroup separation on the basis of diabetes status contributes to higher model accuracies than increasing model complexity when analysing the relationship between blood glucose concentration and insulin concentration. While higher-order polynomial models yielded negligible improvements when applied to the global dataset, separation by diabetic status produced noticeably improved predictive performance, particularly within the non-diabetic cohort as can be seen when comparing R^2 values. This supports the hypothesis that underlying biological complexity between physiological variables and processes may obscure meaningful statistical relationships when considering diabetic and non-diabetic populations as a global dataset.

The broader residual dispersion observed within the diabetic subgroup *(Graph 8)* likely stems from increased metabolic variability associated with insulin resistance and impaired glucose regulation. *Kahn et al. (2006)* describe Type 2 diabetes as a condition involving multiple overlapping defects in insulin action and β-cell responsiveness, which may explain the increasing heteroscedastic behaviour and decrease in predictive performance of the linear model at higher blood glucose concentrations. Similarly, the inflammatory pathways associated with diabetic progression discussed by *Shoelson et al. (2006)* may contribute to greater physiological unpredictability and higher variability in insulin concentrations.

The comparatively stronger model fit observed within the non-diabetic subgroup suggests that insulin response under normal metabolic regulation follows a more predictive relationship with blood glucose concentration. This aligns with findings by *Stumvoll et al. (2005)*, who note that early disruptions in insulin sensitivity often precede the greater metabolic instability characteristic of established diabetic pathology.

Citations

1) Khare, A. D. (n.d.). *Diabetes dataset*. Kaggle.
2) Ramachandran, A., & Snehalatha, C. (2009). Current scenario of diabetes in India. *Journal of Diabetes, 1*(1), 18–28. https://doi.org/10.1111/j.1753-0407.2008.00004.x
3) Astrup, A., & Finer, N. (2000). Obesity and Type 2 diabetes: A growing challenge. *British Journal of Nutrition, 83*(S1), S113–S120.
4) Stumvoll, M., Goldstein, B. J., & van Haeften, T. W. (2005). Type 2 diabetes: Principles of pathogenesis and therapy. *The Lancet, 365*(9467), 1333–1346.
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